

Working with Namespaces:

```
kubectl create namespace demo
```

```
kubectl get namespaces
```

```
kubectl get pods -n demo
```

```
apiVersion: v1
```

```
kind: Pod
```

```
metadata:
```

```
  name: nginx
```

```
  namespace: demo
```

```
spec:
```

```
  containers:
```

```
    - name: nginx
```

```
      image: nginx
```

```
kubectl config set-context --current --namespace=demo
```

```
kubectl delete namespace demo
```

Advanced Use cases:

```
apiVersion: v1
```

```
kind: ResourceQuota
```

```
metadata:
```

```
  name: demo-quota
```

```
  namespace: demo
```

```
spec:
```

```
  hard:
```

```
requests.cpu: "2"
requests.memory: 2Gi
limits.cpu: "4"
limits.memory: 4Gi
```

```
apiVersion: rbac.authorization.k8s.io/v1
kind: Role
metadata:
  namespace: demo
  name: read-only
rules:
- apiGroups: [""]
  resources: ["pods"]
  verbs: ["get", "list"]
```

Label selectors:

```
kubectl get pods -l app=frontend
```

```
kubectl get pods -l app!=backend
```

```
kubectl get pods -l 'env in (dev,prod)'
```

```
kubectl get pods -l 'app'
```

Using Label Selectors

```
selector:
  matchExpressions:
  - key: env
```

operator: In

values:

- prod
- preprod

selector:

matchLabels:

app: nginx

version: v2

podSelector:

matchExpressions:

- key: role

operator: In

values:

- db
- cache

Annotation versus Labels

apiVersion: v1

kind: Pod

metadata:

name: nginx

namespace: demo

annotations:

description: "NGINX app"

createdBy: "Demo team"

prometheus.io: "true"

spec:

containers:

- name: nginx
- image: nginx

Restart policy types:

restartPolicy: Always

restartPolicy: onFailure

restartPolicy: Never

Deployment Overview

apiVersion: apps/v1

kind: Deployment

metadata:

- name: nginx-deployment

- labels:

 - app: nginx

spec:

- replicas: 3

- selector:

 - matchLabels:

 - app: nginx

template:

- metadata:

 - labels:

 - app: nginx

- spec:

 - containers:

 - name: nginx

 - image: nginx:1.26.3

ports:

- containerPort: 80

Rolling updates:

strategy:

- type: RollingUpdate

- rollingUpdate:

 - maxUnavailable: 1

 - maxSurge: 1

kubectl set image deployment/nginx nginx=nginx:1.28.0

kubectl rollout status deployment/nginx

Rollbacks

kubectl rollout history deployment/nginx

kubectl rollout undo deployment/nginx

kubectl rollout undo deployment/nginx --to=revision=2

Scaling deployments:

kubectl scale deployment nginx-deployment --replicas=2

spec:

- replicas: 3

kubectl patch deployment nginx-deployment -p '{"spec": {"replicas": 4}}'

```
kubectl get deployment nginx-deployment
```

```
kubectl autoscale deployment nginx-deployment --min=2 --max=5 --cpu-percent=70
```

StatefulSets:

```
apiVersion: apps/v1
```

```
kind: StatefulSet
```

```
metadata:
```

```
  name: webapp
```

```
spec:
```

```
  serviceName: "webapp"
```

```
  replicas: 3
```

```
  selector:
```

```
    matchLabels:
```

```
      app: webapp
```

```
  template:
```

```
    metadata:
```

```
      labels:
```

```
        app: webapp
```

```
    spec:
```

```
      containers:
```

```
        - name: nginx
```

```
          image: nginx
```

```
          volumeMounts:
```

```
            - name: www
```

```
              mountPath: /usr/share/nginx/html
```

```
  volumeClaimTemplates:
```

```
    - metadata:
```

```
      name: www
```

```
    spec:
```

```
      accessModes: ["ReadWriteOnce"]
```

resources:
requests:
storage: 1Gi

DaemonSets:

apiVersion: apps/v1
kind: DaemonSet
metadata:
 name: logagent
spec:
 selector:
 matchLabels:
 name: logagent
 template:
 metadata:
 labels:
 name: logagent
 spec:
 containers:
 - name: logagent
 image: fluentd

Update Strategies for DaemonSets

updateStrategy:
 type: OnDelete

updateStrategy:
 type: RollingUpdate
 rollingUpdate:
 maxUnavailable: 1

Overview of Jobs

apiVersion: batch/v1

kind: Job

metadata:

name: hellojob

spec:

template:

spec:

containers:

- name: hello

image: busybox

command: ["echo", "Hello from a Kubernetes Job"]

restartPolicy: Never

backoffLimit: 3

CronJobs basics

apiVersion: batch/v1

kind: CronJob

metadata:

name: hellocron

spec:

schedule: "*/10 * * * *" # Every 10 minutes

jobTemplate:

spec:

template:

spec:

containers:

- name: hello

image: busybox

command: ["echo", "Hello from CronJob"]

restartPolicy: OnFailure

